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(See 6.5)

MILITARY SPECIFICATION

INDICATORS, COMBUSTIBLE GAS AND OXYGEN, PORTABLE

This limited coordination Military Specification has been prepared by the Naval Ship Engineering Center based upon currently available technical information, but it has not been approved for promulgation as a coordinated revision of Military Specification MIL-I-2703B. It is subject to modification. However, pending its promulgation as a coordinated Military specification, it may be used in procurement.

1. SCOPE

1.1 Scope. This specification covers portable indicators used for detecting and indicating the concentration of combustible gases and vapors in air and for indicating the oxygen content in air (see 6.2).

1.2 Classification. Portable combustible gas and oxygen indicators shall be of the following types and classes as specified (see 6.1.1):

Class 1 - Combustible gas, types A or B.
Class 2 - Oxygen.
Class 3 - Dual units - combustible gas and oxygen.

Type A - Detect miscellaneous combustible gases or vapors, and atmospheres containing nonleaded gasoline vapors (see 3.4.1.1 and 3.4.1.4).
Type B - Detect miscellaneous combustible gases or vapors, including leaded gasoline vapors (see 3.4.1.2).

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL
ZZ-H-461 - Hose, Rubber, Gas (Acetylene-Hydrogen, Air and Oxygen).

MILITARY
MIL-R-94 - Resistor, Variable, Composition, General Specification For.
MIL-B-117 - Bag, Sleeve and Tubing - Interior Packaging.
MIL-W-530 - Webbing, Textile, Cotton, General Purpose, Natural or in Colors.
MIL-I-983 - Interior Communication Equipment, Naval Shipboard; Basic Design Requirements for.
MIL-L-3122 - Leather, Cattlehide, For Footwear Uppers, Chrome Tanned, Fat, Liquored.
MIL-S-3786 - Switches, Rotary (Circuit Selector, Low-Current Capacity), General Specification For.
MIL-S-3950 - Switch, Toggle, Environmentally Sealed, General Specification for.
MIL-M-7298 - Manual, Technical, Commercial Equipment.
MIL-S-8805 - Switches and Switch Assemblies, Sensitive and Push (Snap Action), General Specification For.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Ship Engineering Center, SEC 6124, Department of the Navy, Washington, D.C. 20362 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MILITARY (Continued)

MIL-M-10304 - Meters, Electrical Indicating, Panel Type, Ruggedized, General Specification For.
MIL-P-15024 - Plates, Tags and Bands for Identification of Equipment.
MIL-P-15024/5 - Plates, Identification.
MIL-M-15071 - Manuals, Technical: Equipments and Systems, Content Requirements for.
MIL-M-16034 - Meter, Electrical - Indicating (Switchboard and Portable Types).
MIL-E-17555 - Electronic and Electrical Equipment, Accessories and Repair Parts, Packaging and Packing of.
MIL-S-18396 - Switches, Meter and Control, Naval Shipboard.
MIL-B-55130 - Battery, Storage, Sealed, Nickel Cadmium.
MIL-B-81847 - Bag, Flexible, Electrostatic-free, Watervaporproof, Heat Sealable.

STANDARDS

FEDERAL

FED-STD-313 - Symbol for Packages and Containers for Hazardous Industrial Chemicals and Materials.

MILITARY

MIL-STD-108 - Definition of and Basic Requirement for Enclosure for Electric and Electronic Equipment.
MIL-STD-167-1 - Mechanical Vibrations of Shipboard Equipment (Type I - Environmental and Type II - Internally Excited).
MIL-STD-454 - Standard General Requirements for Electronic Equipment.
MIL-STD-810 - Environmental Test Methods.
MIL-STD-831 - Test Reports, Preparation of.
MIL-STD-1399 - Interface Standard For Shipboard Systems.
Section 103 - Interface Standard For Shipboard Systems, Electric Power Alternating Current.
MS15011 - Fittings, Hose, Welding and Cutting, Oxygen and Acetylene.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

BUREAU OF MINES

Bulletin 503 - Limits of Flammability of Gases and Vapors.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D.C. 20002.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

National Fire Codes - Vol. 13 - 1976 - Manual 325M - 1969 - Fire-Hazard, Properties, Flammable Liquids, Gases and Volatile Solids.

(Application for copies should be addressed to National Fire Protection Association, 60 Batterymarch Street, Boston, Massachusetts 02110.)

UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification Rules.

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Drive, Chicago, Illinois 60606.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Qualification. Indicators furnished under this specification shall be products which are qualified for listing on the applicable qualified products list at the time set for opening of bids (see 4.3 and 6.3).

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3.2 Material. Material shall be selected from MIL-I-983 or, shall be approved by the Naval Ship Engineering Center (NAVSEC) as specified in MIL-I-983. Even though constituent material is approved, finished products shall conform to all requirements of this specification and shall perform their intended function.

3.2.1 Selection of parts. Except as otherwise specified herein, the parts shall be selected from those listed in MIL-I-983.

3.2.2 Electrical circuits. The electrical circuits shall be designed with solid-state parts and shall not contain electron tubes or relays.

3.2.3 Meters. Electrical indicating meters shall be high-impact shock resistant, watertight, or sealed types in accordance with the applicable requirements of MIL-M-10304 or MIL-M-16034.

3.2.4 Switches and controls. The switches and controls shall conform to the following:

(a) Meter and control switches - MIL-S-18396.
(b) Pushbutton switches - MIL-S-8805.
(c) Toggle switches - MIL-S-3950.
(d) Variable resistors - MIL-R-94.

3.2.5 Charging cable. The battery charging cable shall be a three-wire cable assembly, 10 feet long. One end shall be fitted with a grounded type male connector, to mate with a standard bulkhead receptacle. The other end shall be fitted with a grounded type female connector, to mate with the connector on the indicator enclosure (see 3.3.3.1).

3.3 Design and construction.

3.3.1 Description. Each indicator shall include an indicating meter (or meters for class 3 units); adjustment and operating controls and switch(es); sensing heads, an integral, rechargeable power supply; a battery charging circuit and indicating lamp; and an electrically driven pump. Each indicator, single or dual, shall be self-contained and shall be assembled as one unit.

3.3.2 Enclosure. A hinged top cover shall be furnished and the enclosure shall be splashproof in accordance with MIL-STD-108 with the cover closed. The top surface (under the cover) of the enclosure shall include the indicating meter or meters, meter adjustment and operating controls and switch(es) and battery charger circuit and indicating lamp, all conveniently located for operation.

3.3.2.1 A sampling hose or probe tube connection and a battery charging cable connection shall be made an integral part of the enclosure (see 3.3.3.1 and 3.3.3.2).

3.3.2.2 Operating instructions. Operating instructions shall be supplied with each enclosure such that it may be readily referred to while operating the indicator. The instructions shall also contain specific equipment limitations and safety information (see 3.3.4.1.1).

3.3.2.3 Carrying strap. A carrying strap (see 3.8.11) shall be furnished to permit the indicator to be hung about the neck, leaving the hands free to operate the controls.

3.3.2.4 All units shall have printed operating/response check instructions attached. Class 1 and class 3 units shall also have an attached chart or charts showing a set of curves to interpret the meter indication for each type of detectable gas. These instructions and charts shall be protected by laminated plastic as approved by NAVSEC.

3.3.2.5 The class 1 and class 3 units shall also contain provisions for storage of two spare combustible filament(s) or sensing heads (see 3.3.7.1 and 3.6.4).

3.3.3 Connections.

3.3.3.1 Charging cable (electrical). The side of the enclosure shall contain a recessed electrical connector (male) for internal battery charging. The connector shall mate with the connector on the electrical charging cable furnished (see 3.2.5), and shall contain a self-closing cover or a threaded cap permanently affixed to the enclosure.

3.3.3.2 Sampling hose. The enclosure shall contain an air-tight fitting for attachment of the sampling hose. The hose connection shall be male threaded. The threaded portion, hole and countersunk opening shall be similar to that shown for hose coupling (oxygen), piece 2, of MS15011. The countersunk opening shall be arranged to make an airtight coupling with the coupling end of the sampling hose connection.

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3.3.4 Power supply. The power supply shall consist of a leak-proof, rechargeable nickel cadmium storage battery conforming to MIL-B-55130. The nominal voltage rating of the battery shall be 6 volts (V) or less. The capacity shall be such to provide sufficient power to operate the indicator for a minimum of 8 hours from a full charge condition. The unit shall be capable of operating from 105 to 120 volts alternating current (Vac) regardless of battery charge condition. Each unit shall contain the necessary controls and electrical circuitry for checking the charge condition of the battery on the indicator meter.

3.3.4.1 An integral battery charging circuit, capable of operating from a 105 to 120 volt 60 Hertz (Hz) single phase alternating current (a.c.) source as specified in section 103 of MIL-STD-1399 shall be provided. The charging circuit shall not damage the batteries when the battery is fully charged and the charging circuit remains energized. The battery charging indicator ramp shall be energized and remain energized during charging of the battery.

3.3.4.1.1 The leakage current of the indicator shall not exceed 5 milliamperes (mA) to ground with the battery charging circuit in operation. On equipments which have an unavoidable leakage current in excess of 5 mA, a warning plate shall be attached to the front panel reading:

"DANGER - Do not energize this equipment unless frame and all exposed metal parts are grounded."

3.3.5 Atmosphere sampling. The atmosphere under test shall be drawn through the sensing head(s) of the indicator by means of an integral, battery driven pump (see 3.3.5.1). The pump shall be capable of drawing a continuous sample of the atmosphere through 30 feet of sampling hose with a 3-foot probe tube attached. Instrument response time, based on sample transit time shall be noted for this mode of operation (see 3.6.7).

3.3.5.1 Class 3 units. The class 3 units shall have a common sample inlet and when activated, the pump shall draw the atmosphere through bolt sensing heads. The meter and meter circuits associated with each indicator shall be separate, individual circuits, with no electrical or mechanical interaction other than the common power supply.

3.3.6 Voltage adjustment and zero setting.

3.3.6.1 Class 1 unit. Class 1 indicators shall contain the necessary controls and electrical circuits to adjust the voltage impressed upon the terminals of the bridge network to zero the meter or, provide regulated voltage and means of zeroing the meter.

3.3.6.2 Class 2 unit. The class 2 indicators shall contain the necessary controls and electrical circuits to adjust the voltage for the correct meter indication during calibration.

3.3.6.3 Class 3 unit. Class 3 indicators shall contain the controls and electrical circuits specified for both the class 1 and class 2 indicators.

3.3.7 Detection means.

3.3.7.1 Class 1 and 3 units. Class 1 and 3 indicators shall contain a filament bridge type resistive network to detect the presence of combustible gases and vapors in air.

3.3.7.2 Class 2 units. Class 2 indicators shall contain a leak-proof, self-generating, electrolytic cell to measure the oxygen content in air.

3.3.8 Response check. The indicator design shall facilitate provisions for response check without requiring disassembly of the unit and without requiring external indicators or special fittings.

3.3.8.1 Response check units. All class 1 and 3 indicators provided under this specification shall be capable of response check using propane in air mixture of 2.2 percent. Response check curves shall be provided as required in 3.5.2.5.

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3.3.9 Dimensions and weight. The indicators shall not exceed the dimensions and weight shown in table I.

TABLE I. Dimensions and weight.

Indicator	Inches, maximum			Pounds, maximum
	Height	Width	Depth	Weight
Class 1 (combustible gas)	8	7-1/2	3-1/2	10
Class 2 (oxygen)	8	6	5	7-1/2
Class 3 (dual unit)	8	11-1/2	7	15

3.4 Performance.

3.4.1 Operation.

3.4.1.1 Class 1, type A. The class 1, type A indicator shall be capable of safely and accurately detecting the miscellaneous combustible gases and vapors associated with fuel oils or gasoline without tetraethyl lead, paints, alcohol, and acetone vapors, fuel gas, hydrogen, and acetylene in mixtures with air or oxygen. The indicators shall give a reliable indication to show when the mixture is in or above the explosive range or below the lower explosive limit (LEL) shown in table II, except as otherwise specified. Where no LEL of vapors or gases is specified, the limits given in Bureau of Mines Bulletin 503 and National Fire Protection Association Manual 325M shall be used as guidelines.

TABLE II. Lower explosive limit for indicators.

Liquid gas and vapor	LEL in air (percent by volume)
Ethyl alcohol (test without sampling hose)	3.3
Hydrogen	4.0
Benzene	1.4
Methane	5.3
Leaded gasoline with octane rating of 80 and above[1/]	1.3
Hexane	1.25
Acetylene	2.3
Acetone	2.55
Pentane	1.4
Jet and reciprocating engine fuels	
JP-4	1.4
JP-5	0.60
JP-8	0.72
100/130 octane aviation gasoline[1/]	1.3

[1/] Applicable to type B only.

3.4.1.2 Class 1, type B. The class 1, type B indicator shall be capable of safely and accurately detecting combustible gases or vapors in mixtures with air or oxygen. The indicator shall give a reliable indication to show when the mixture is in or above the explosive range or below the LEL (see table II).

3.4.1.3 Class 2. The class 2 indicator shall be capable of measuring the oxygen content of atmospheres in the range of 1 to 25 percent by volume. The class 2 indicator shall be self-calibrating.

3.4.1.4 Class 3. The class 3 unit shall contain a class 1, type A, and class 2 indicator.

3.4.2 Operating controls. The meter adjusting control knob(s) or switch (or both) for obtaining the initial zero or response check setting of the indicating meter or meters shall have sufficient tension or shall be provided with a disengaging clutch in order to ensure that the meter setting cannot be accidentally disturbed. When a separate variable resistor is used for exclusive use of controlling voltage and the correct operating voltage is indicated on the indicating meter scale, a clutch type disengaging control knob need not be used. The meter adjusting control shall also provide a means for manually turning the

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power on and off. On class 3 units, this feature shall be incorporated in only one control. A switch or control may be provided to automatically turn the power off when the top cover is closed.

3.4.2.1 The battery check (test) switch or control shall be of the momentary make type and shall be normally open.

3.4.2.2 The instrument power switch shall also energize the integral sampling pump (see 3.3.5).

3.4.2.3 Class 1, 2, and 3 indicator meters shall contain a means of providing a continuous indication of the highest measurement received during sampling, (that is, meter pointer lock, meter pointer follower, or other means). A means for manual resetting shall be provided.

3.4.3 Calibration.

3.4.3.1 Class 1 units. Detail calibration of the class 1 indicators shall be accomplished by a laboratory using samples of controlled concentrations. Shipboard response checks shall be as specified in 3.3.8.1.

3.4.3.2 Class 2 unit. The class 2 indicator shall be capable of self-calibration in uncontaminated air.

3.4.4 Scale and accuracy.

3.4.4.1 Class 1 units. The scale of the meter shall be graduated from 0 to 100 percent of the LEL and shall contain 20 or more uniform divisions. The portion above the 60 percent mark shall be colored red. The accuracy of the indication shall be such that the value read from the supplied curve chart shall be within plus or minus 20 percent of the true value of an applied known mixture.

3.4.4.2 Class 2 units. The scale of the meter shall be graduated from 0 to 25 percent and shall contain 26 or more uniform divisions. The accuracy of the indication shall be plus or minus 5 percent of full scale (see 4.5.2.1).

3.4.5 Filaments. The filaments of the class 1, type A and B indicators shall not be affected by operation in hydrogen sulfide when subjected to the tests specified in 4.5.2.3. The filaments shall be operated at a sufficiently high temperature to prevent poisoning by the gases specified in table II and shall not be affected when subjected to the tests specified in 4.5.2.4.

3.4.6 Stability. The meter indications shall vary not more than plus or minus 5 percent of the full scale value during the test specified in 4.5.2.2.

3.4.7 Response time. The indicator shall provide a stable reading within 30 seconds when subjected to the tests specified in 4.5.2.5. With 30-foot sampling hose attached, response time shall be as specified in instructions attached to the unit.

3.4.8 Battery voltage indicator. The meter scale shall contain distinctive markings to provide a GO/NO-GO type indication of the battery condition.

3.4.9 Battery charge cycle. The battery charging circuitry shall be capable of returning the battery to its full charge voltage within 16 hours or less from a 90 percent discharge condition.

3.5 Environmental.

3.5.1 Temperature. The indicators shall conform to the accuracy requirements of 3.4.4 over the ambient temperature range of 0 degrees Celsius (°C) to 50°C when tested as specified in 4.5.3 and 4.5.4.

3.5.2 Humidity. The indicators shall conform to the accuracy requirements of 3.4.4 during and after subjection to a relative humidity of 95 percent (see 4.5.5).

3.5.3 Inclination. The accuracy of operation of the indicator shall not be adversely affected when the indicator is inclined up to 30 degrees from the normal operating position (see 4.5.6).

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3.5.4 Vibration. The indicators shall conform to the accuracy, stability, and response time requirements and there shall be no evidence of mechanical damage, loosening of parts or change in status of operating control after subjection to the type I vibration test of MIL-STD-167-1.

3.5.5 Salt fog. The indicators shall conform to the accuracy requirements of 3.4.4 after subjection to the salt fog test of 4.5.8.

3.5.6 Life.

3.5.6.1 Filament. The class 1 filament shall be operable and shall meet the accuracy requirements of 3.6.4 after the life tests specified in 4.5.10.

3.5.6.2 Oxygen cell (class 2). The oxygen cell shall be capable of operation for a minimum of 1 year (see 4.5.10.2).

3.5.6.3 Battery. The battery shall be capable of operating the indicator without degradation of indication for a continuous period of 8 hours before requiring recharging.

3.5.6.4 Unit. The unit with the exception of the pump which shall be not less than 2000 hours, shall be capable of operating not less than 10,000 hours subject to periodic replacement of filaments in the class 1 indicators, replacement of oxygen cell in the class 2 indicator, and recharging the battery as required. Lamp replacement will be allowed.

3.5.7 Flashback protection. The chamber housing the filaments of both types of combustible gas indicators shall be provided with flame arrestors in order to prevent flashbacks from igniting outside gases. The indicator shall pass the test specified in 4.5.11 without ignition of the gases.

3.6 Detail requirements. Detail requirements applicable to individual, specific classes and tubes of indicators shall be as specified on the applicable approved manufacturer's drawing (see 3.8).

3.6.1 Sampling hose. The flexible sampling hose shall be of non-absorbent fabric reinforced rubber, impervious to petroleum products, benzene, and water vapor in accordance with ZZ-H-461. The ends of each length of hose shall be fitted with airtight fittings as shown on MS15011. One end of the hose shall be fitted with a hose connection gland (straight), piece 5 of MS15011, holding captive, loosely, a threaded connection nut (oxygen), piece 4 of MS15011 which shall be used as a hose connection gland. The other end of the hose shall be fitted with a hose connection fitting, piece 4, except that it shall be arranged with a male threaded connection; the threaded portion, hole and countersunk opening shown for hose coupling (oxygen), piece 2 of MS15011. The countersunk opening shall be arranged to make an airtight coupling with the coupling end of a fitting to hose connection gland (straight), piece 5 of MS15011.

3.6.1.1 Hose length. Unless otherwise specified (see 6.1.1), two hoses, each 15 feet long shall be furnished. When specified (see 6.1.1), a 5-foot length of hose shall be furnished with each indicator either as additional or as a substitute for one of the 15-foot length hose.

3.6.2 Probe tube. When specified (see 6.1.1), a probe tube of brass or aluminum tubing 3 feet long shall be furnished. One end of the probe tube shall be fitted with a female airtight fitting for attachment to the male threaded portion of the sampling hose or the fitting of the indicator enclosure. Fittings should be arranged in a similar manner to the arrangement used on the sampling hose by holding captive, loosely, a connection nut (oxygen), piece 4 of MS15011, which shall be used as a hose connection gland.

3.6.3 Remote operation. The class 1 indicators shall be capable of continuously sampling gas or vapor mixtures without manual operation by the operator, at distances up to at least 30 feet from the indicator. Continuous operation shall be possible from an external 115 V, 60 Hz source, while maintaining the battery at full charge.

3.6.4 Filaments or sensing heads. The metallic filaments used as catalytic reagents in the bridge type resistive network shall be individually replaceable or, replaceable as part of the sensing head containing the filament. Replacement shall be by means of a plug-in, screw type or similar method. A spare combustible filament or sensing head(s) shall be supplied with each class 1 and class 3 indicator.

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3.6.5 Oxygen cell (class 2). The oxygen cell shall be a plug-in or screw-in type of self-generating electrolytic cell that produces a current directly proportional to the amount of oxygen sensed.

3.6.6 Carrying strap. The carrying strap shall be fabricated of either cotton webbing conforming to type III, Olive Drab No. 7 of MIL-W-530, or chrome leather conforming to MIL-L-3122. Either material shall be treated for mildew resistance.

3.6.7 Response check kit. The class 1 indicator response check kit shall contain a 2.2 percent propane in air mixture packaged such to allow preparation of test samples which will produce approximately 1/4 and 1/2 scale deflections.

3.7 Identification marking. The indicator unit shall include an identification plate with complete information as specified in MIL-P-15024/5. Equipment assemblies and parts shall be marked for identification in accordance with MIL-I-983.

3.8 Technical data. The contractor shall prepare engineering drawings and technical manuals in accordance with the data ordering documents included in the contract (see 6.1.2).

3.8.1 Drawings. In addition, the following unique technical features shall be included:

(a) Classification of equipment (see 1.2).
(b) Manufacturer's name and identification number of equipment.
(c) List of descriptive data including:
    (1) Applicable specification and exceptions.
    (2) Complete rating showing battery voltage, battery charging input voltage and frequency, outputs, and others.
    (3) Enclosure - size, weight.
    (4) Types of combustible gases detectable (class 1 and 3).
    (5) Notation of the corrosion resistant treatment of all hardware.
    (6) Special features.
(d) A complete wiring diagram and a complete schematic circuit.
(e) Finish, including method of treatment of enclosure for painting, color, and applicable specification of paint.
(f) List of material (excluding items of hardware).
(g) List of repair parts and special tools, in table form.
(h) Outline of equipment showing overall and principle dimensions of front view, side view, top view, and sectional views, as necessary, to show mounting arrangement and installation details and location of parts in the list of material. Parts shall be flagged by item or piece number. Views shall show identification plates with data to appear thereon.
(i) A table or list of parts such as printed circuit boards, variable resistors, capacitors, and similar items giving piece number, type, and rating of each. This data is a supplement to the list of material where space normally does not permit complete listing of part rating. Ratings shall include both manufacturer's rating and the specific application rating.

3.8.2 Manuals. Technical manuals shall be prepared in accordance with type I, commercial, of MIL-M-15071 or MIL-M-7298. When MIL-M-15071 is specified, the content and arrangement shall be modified as follows:

(a) Front matter.
(b) General information.
(c) Operation - This section shall contain operating and calibration procedures to include a chart showing a set of curves to interpret the meter indications for each type combustible gas indicator.
(d) Functional description.
(e) Maintenance, repair, and troubleshooting.
(f) Parts list.

3.8.2.1 Technical manual validation. NAVSEC technical representative shall receive 20 days prior notice of validation so that NAVSEC may have the opportunity to witness the validation demonstration if desired. The validation shall be done continuously without interruption during regular working hours until complete. Operation, maintenance, refurbishing, and calibration procedure validity and completeness shall be demonstrated. Omission and error shall be corrected by the contractor before final manual approval.

3.9 Repair parts. Repair parts shall be as specified (see 6.1.1).

3.10 Workmanship. The workmanship on the indicators shall be in accordance with requirement 9 of MIL-STD-454.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 The contractor shall provide and maintain an inspection system in accordance with the data ordering documents included in the contract (see 6.1.2).

4.1.2 Test conditions. Unless otherwise specified (see 6.1.1), tests shall be made under the following conditions:

(a) Ambient temperature. Tests shall be performed at an ambient temperature of 25°C ± 15°C.
(b) Ambient altitude. Tests shall be run at approximately sea-level altitude. Test data shall include the dewpoint at which all altitude tests are conducted.
(c) Tolerance and accuracy. The tolerance of test conditions, other than ambient, and the accuracy of test apparatus shall be as specified in MIL-STD-810.
(d) Attitude. Unless otherwise specified (see 6.1.1), the indicator shall be tested in the normal operating position.

4.2 Classification of inspections. The methods of examination and testing shall be as follows:

(a) Qualification inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

4.3 Qualification tests. Qualification tests shall be conducted at a laboratory satisfactory to the Naval Ship Engineering Center. Qualification tests shall consist of the examination and tests shown in table III.

TABLE III. Qualification examinations and tests.

Examination and tests	Unit sample number		Requirement paragraph	Test paragraph
	1	2		
General examination	x	x	3.2, 3.3, and 3.6	4.5.1
Calibration and accuracy	x	x	3.4.3 and 3.4.4	4.5.2.1
Accuracy of indication	x	x	3.4.4.1 and 3.4.4.2	4.5.2.1.1 1/ and 4.5.2.1.2 2/
Response time	x	x	3.4.7	4.5.2.5
Temperature	x	x	3.5.1	4.5.3 and 4.5.4
Humidity	x		3.5.2	4.5.5
Inclination	x		3.5.3	4.5.6
Vibration	x		3.5.4	4.5.9
Salt fog	x		3.5.5	4.5.8
Hydrogen sulfide 1/		x	3.4.5	4.5.2.3
Life		x	3.5.6	4.5.10
Battery		x	3.5.6.3	4.5.10.1 1/ and 4.5.10.2 2/
Filament 1/		x	3.5.6.1	4.5.10.1
Oxygen cell 2/		x	3.5.6.2	4.5.10.2
Battery charge rate		x	3.4.9 and 3.4.10	4.5.2.3
Tetraethyl lead 1/		x	3.4.5	4.5.2.4
Explosion proof (safety) 1/		x	3.5.7	4.5.11
Enclosure		x	3.3.2	4.5.7
Stability	x	x	3.4.6	4.5.2.2

1/ Class 1 indicator only.
2/ Class 2 indicator only.

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4.3.1 Qualification sample. Two indicators of each class and type for which qualification is sought shall be submitted for test. A response check kit shall also be furnished for the tests specified in 4.5.2.1.1 (see 3.6.7).

4.3.2 Tests. Equipments submitted for qualification inspection shall be subjected to the tests shown in table III in the order listed. Failure of an equipment to comply with any of the requirements listed may be cause for refusal to grant qualification.

4.3.3 Data submission - drawings and inspections. Together with the test samples, the manufacturer shall submit one reproducible copy of the drawings (see 3.8.1) and two printed copies of simple instructions with illustrations and diagrams, if necessary, covering the operation of the indicator.

4.3.4 Disposition of qualification samples and test report. Upon completion of the qualification tests, the test samples shall be identified as required and forwarded to NAVSEC (see 6.3). The test samples shall be accompanied with three copies of the qualification test report, prepared in accordance with MIL-STD-831.

4.3.5 Retention of qualification. To retain qualification, the contractor shall comply with 4.4.2.3 and the group C inspection of 4.4.3. In the event that no production occurred during a 36-month period, a report shall be submitted certifying that the company still has the capabilities and facilities necessary to produce the item. If during two consecutive periods there has been no production, the manufacturer may be required, at the discretion of NAVSEC, to submit a representative product for testing in accordance with the qualification inspection requirements.

4.4 Quality conformance inspection. Quality conformance inspection shall consist of the following groups of examinations and tests:

Group A (see 4.4.2.1).
Group B (see 4.4.2.2).
Group C (see 4.4.2.3).

4.4.1 Inspection lot. All indicators of the same class and type presented for quality conformance inspection at one time shall be considered a lot. The lot may include the entire contract quantity or it may be the production of any convenient time period.

4.4.2 Sampling plan.

4.4.2.1 Sampling for group A examination and tests. A sample of indicators shall be selected from each lot in accordance with table IV and subjected to each of the group A examinations and tests specified in 4.4.3, in the order shown.

TABLE IV. Sampling for group A examination tests.

Number of equipments in lot	Number of equipments in sample	(Number of equipments nonconforming on any group A test)	
		Acceptance number	Rejection number
7 and under	All	-	-
8 to 15	7	0	1
16 to 40	10	0	1
41 to 110	15	0	1
111 to 300	25	1	2
301 to 500	35	1	2
501 and over	50	2	3

4.4.2.2 Sampling for group B tests. A sample of indicators shall be selected from each lot in accordance with table V and subjected to each of the group B examinations and tests specified in 4.4.3, in the order shown. Group B tests shall be performed on sample units that have passed the group A tests.

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TABLE V. Sampling for group B tests.

Number of equipments in lot	Number of equipments in sample	(Number of equipments nonconforming on any group B test)	
		Acceptance number	Rejection number
3 and under	All	-	-
4 to 15	3	0	1
16 to 40	5	0	1
41 to 110	7	0	1
111 to 300	10	0	1
301 to 500	15	1	2
501 and over	25	2	3

4.4.2.3 Sampling for group C inspection. Every three years, two indicators of each class and type produced during the period shall be subjected to the group C inspections specified in 4.4.3. The indicators selected shall be sample units that have passed the group A and group B tests. Group C tests will also be required when the basic design of the indicators or the process or material has been changed.

4.4.3 Examination and test. The groups A, B, and C examination and tests of table VI shall be performed in the order listed.

TABLE VI. Quality conformance inspection.

Examination or test	Unit sample number[1]		Requirement paragraph	Test paragraph
	1	2		
Group A				
General examination			3.2, 3.3, and 3.6	4.5.1
Group B				
Calibration and accuracy			3.4.3 and 3.4.4	4.5.2.1
Accuracy of indication			3.4.4.1 and 3.4.4.2	4.5.2.1.1[2] and 4.5.2.1.2[3]
Response time			3.4.7	4.5.2.5
Group C				
Temperature	x	x	3.5.1	4.5.3 and 4.5.4
Humidity	x		3.5.2	4.5.5
Inclination	x		3.5.3	4.5.6
Hydrogen sulfide[2]		x	3.4.5	4.5.2.3
Tetraethyl lead[2]		x	3.4.5	4.5.2.4
Explosion proof (Safety)[2]		x	3.5.7	4.5.11
Life		x	3.5.6	4.5.10
Battery		x	3.5.6.3	4.5.10.1[2] and 4.5.10.2[3]
Filament[2]		x	3.5.6.1	4.5.10.1
Oxygen cell[3]		x	3.5.6.2	4.5.10.2
Battery charge rate		x	3.4.9 and 3.4.10	4.5.2.3
Vibration	x		3.5.4	4.5.9
Salt fog	x		3.5.5	4.5.8
Enclosure (splashproof)		x	3.3.2	4.5.7
Stability	x	x	3.4.6	4.5.2.2

[1] See 4.4.2.3.

[2] Class 1 indicator only.

[3] Class 2 indicator only.

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4.4.3.1 Rejected lots (group A and B inspection). The results of each examination and test shall be compared with the requirements of this specification. Failure to conform to the requirements of this specification for any group A or B test shall be counted as a defect and the equipment shall be rejected. When the number of such nonconforming equipments in any sample exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected.

4.4.3.2 Rejected lot (group C inspection). (Applicable only to units of the contract under which group C tests are being accomplished). The results of each group C test shall be compared with the requirements of this specification. In the event of a failure to conform to the requirements of this specification for any group C test, the contractor shall take corrective action on the materials or processes, or both, as warranted, and on all units of product which can be corrected and which were manufactured under essentially the same conditions, with essentially the same materials, processes, and so forth, and which are considered subject to the same failure. Acceptance of the product shall be discontinued until corrective action, acceptable to the Government, has been taken. After corrective action has been taken, group C inspection shall be repeated on additional sample units (all inspections, or the inspection which the original sample failed, at the option of the Government). Groups A and B inspection may be reinstituted; however, final acceptance shall be withheld until the group C reinspection has shown that the corrective action was successful. In the event of failure after reinspection, complete information concerning the failure and the corrective action taken shall be furnished to the contracting officer.

4.4.4 Disposition of test samples. Sample units that have been subjected to and passed the group A and group B tests may be delivered on a contract. Group C test samples shall not be delivered on a contract.

4.5 Visual examination and test procedures.

4.5.1 General examination. Each indicator shall be subjected to a general examination as specified in MIL-I-983. The examination shall verify compliance with the material and design requirements specified in 3.2, 3.3, 3.6, and approved drawings (see 3.8.1).

4.5.2 Operation and performance.

4.5.2.1 Calibration and accuracy. Each indicator under test shall be checked for calibration with 30 feet of sampling hose and when specified, a 3-foot probe tube before and after the completion of the tests specified in 4.5.2.4, 4.5.2.5, 4.5.10, and 4.5.11, inclusive. In addition, the effect of replacing the filaments or sensing heads shall be checked as part of the calibration before the test specified in 4.5.10.

4.5.2.1.1 Class 1 units. The calibration of the class 1, type A or B units shall be initially checked, using the response check kit (see 3.3.8), on such mixtures that the data obtained can be used for comparison with the curve chart furnished with the indicator. Thereafter, several mixtures at approximately 50 percent of the LEL may be used as a basis for calibration checks in subsequent tests. The accuracy of the readout shall be as specified in 3.4.4.1.

4.5.2.1.2 Class 2 units. The calibration of the class 2 indicators shall be initially checked on known concentrations of oxygen in air giving approximately 1/4, 1/2, 3/4, and full scale deflections. The accuracy of the readout shall be as specified in 3.4.4.2.

4.5.2.1.3 Sampling hose. Immediately following calibration the class 1 indicator sampling hose shall be tested to insure no appreciable amount of vapor was absorbed. After purging (using the internal pump) with uncontaminated air for 30 seconds, the class 1 indicator meter readout shall be 0 $\pm$ 5 percent.

4.5.2.2 Stability.

4.5.2.2.1 Class 1 units. Power shall be applied to each class 1 indicator under test for a minimum of 10 seconds before it is balanced on the zero point. The indicator then shall be operated for three consecutive 15-minute periods while burning a combustible gas-air mixture of known concentration and identity. At no time during any 15-minute period shall the variation in readout exceed 5 percent of the full scale value. The indicator shall be quickly and accurately rebalanced in an uncontaminated atmosphere before each 15-minute period.

4.5.2.2.2 Class 2 units. Power shall be applied to each class 2 indicator under test for a minimum of 10 seconds before it is set on the calibration point. The indicator shall be operated for one 15-minute period while sampling an atmosphere containing approximately 25 percent oxygen by volume. At no time during the 15-minute period shall the variation in readout exceed 5 percent of full scale value.

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4.5.2.3 Hydrogen sulfide. Each class 1 indicator under test shall be operated for a period of 15 minutes in a hydrogen sulfide air mixture containing 10 parts per million (p/p) by volume of hydrogen sulfide. The accuracy of the indicator shall not be impaired beyond the accuracy of 4.5.2.1.

4.5.2.4 Tetraethyl lead. (Applicable to class 1, type B only). Each class 1, type B indicator under test shall be operated for a period of 15 minutes in an atmosphere representing less than half scale of gasoline vapors containing tetraethyl lead. The accuracy of the indicator shall not be impaired or the filament poisoned beyond the accuracy of 4.5.2.1.

4.5.2.5 Response time (sensitivity). The time of response, measured during calibration and accuracy tests, shall be not more than that specified in 3.4.7.

4.5.3 High temperature. The high temperature test shall be conducted in accordance with method 501, procedure II of MIL-STD-810. The highest operating temperature shall be 50°C. The highest non-operating temperature shall be 65°C. The readout accuracy during and after the test shall be as specified in 4.5.2.1.

4.5.4 Low temperature. The low temperature test shall be conducted in accordance with method 502, procedure I of MIL-STD-810. The lowest operating temperature shall be 0°C. The readout accuracy during and after the test shall be as specified in 4.5.2.1.

4.5.5 Humidity. The humidity test shall be conducted in accordance with method 507, procedure I, of MIL-STD-810. The highest operating temperature shall be 50°C. The readout accuracy during and after the test shall be as specified in 4.5.2.1.

4.5.6 Inclination. The indicator shall be inclined 30 degrees each side respectively of the normal operating position and remain within the accuracy requirements specified in 3.4.4.

4.5.7 Enclosure. The indicator, with cover closed, shall be subjected to the splash-proof test of MIL-STD-108. After the test, the indicator shall be subjected to the accuracy test of 4.5.2.1.

4.5.8 Salt fog. The salt fog test shall be conducted in accordance with method 509, procedure I, of MIL-STD-810. The indicator shall not be operated during the test. After the test, the indicator shall be subjected to the accuracy test of 4.5.2.1.

4.5.9 Vibration. The indicator shall be tested in accordance with type I of MIL-STD-167-1. During the test, all operating controls shall be observed for a change in status. After vibration test, the indicator shall be subjected to the following examination and tests:

(a) Accuracy (see 4.5.2.1).
(b) Stability (see 4.5.2.2).
(c) Response time (see 4.5.2.5).
(d) Visual examination for evidence of mechanical damage or loosening of parts.

4.5.10 Life.

4.5.10.1 Class 1 units. Each class 1 indicator under test shall be operated for a total of 128 hours at approximately half-scale reading on a mixture of iso-butane or methane and air, in 4 periods of 32 hours each. During each 32 hour period, the indicator shall be operated 8 hours on 120 V, 60 Hz, 8 hours on the battery, and 16 hours with the charging cable connected, and the battery on charge. The battery shall be fully charged at the end of each 16 hour period. During the test and at the end of 128 hours service, the indicator shall be operable and accurate within the limits specified herein.

4.5.10.2 Class 2 units. The class 2 indicators under test shall be operated continuously for a period of 128 hours to determine part reliability. During the test, the indicator shall be operated, alternately, 8 hours on 120 V, 60 Hz, 8 hours on the battery and 16 hours with the charging cable connected and the battery on charge. The battery shall contain a full charge at the end of each 16 hour charging period. During the test and at the end of 128 hours service, the indicator shall be operable and accurate within the limits specified herein.

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4.5.11 Flashback or explosion protection. Each indicator under test shall be checked with oxyhydrogen and oxyacetylene mixtures of maximum explosibility; namely, mixtures of two parts hydrogen to one part oxygen and mixtures of one part acetylene to two and one-half parts oxygen. No flashback or explosion shall occur as a result of these tests.

4.5.12 Leakage current. The leakage current between the power line and indicator shall be measured through a 1500 ohm resistive load at normal line voltage to determine conformance with 3.3.4.1.1.

4.6 Inspection of the preparation for delivery. Sample packages and packs and the inspection of the preservation-packaging, packing, and marking for shipment and storage shall be in accordance with the requirements of section 5 and the documents specified therein.

5. PREPARATION FOR DELIVERY

(The preparation for delivery requirements specified herein apply only for direct Government procurements. For the extent of applicability of the preparation for delivery requirements of referenced documents listed in section 2, see 6.4.)

5.1 Preservation-packaging, packing, and marking. Equipment shall be preserved-packaged level A or C; packed level A, B, or C as specified (see 6.1.1) and marked in accordance with MIL-E-17555, except that the rechargeable battery shall not be removed from the basic equipment and method II unit packaging shall apply for each indicator and its applicable accessories.

5.2 Hazardous materials. Response check kits (see 3.6.7) as specified (see 6.1.1) shall be individually identified in accordance with FED-STD-313. Shipping containers containing hazardous materials shall have a copy of the Safety Data Sheet DD 1813 attached to the shipping document for each destination.

5.3 Repair parts. Repair parts (see 3.9 and 6.1.1) as specified, shall be preserved-packaged, packed, and marked in accordance with MIL-E-17555 as applicable for the intended use and destination as follows:

	Levels	
Destination	Preservation-packaging	Packing
Onboard (accompanying equipment)	A	C
Stock	A	B
Immediate use	C	C

5.4 Semi-conductors or solid-state devices. Repair parts which are semi-conductors or solid-state devices, such as diodes, transistors, integrated circuits, as well as circuit boards or chassis in which they are incorporated, shall be individually packaged in a barrier bag conforming to class 1 of MIL-B-81847, or class E, style 1, type I or II of MIL-B-117. MIL-B-117 bag material shall employ aluminum foil as a laminate of the bag barrier material. Leads and terminals shall be protected from damage by means of the container (carrier) design, die cut inserts, or by the use of non-corrosive cushioning material. Leads and other projecting parts may be used for positioning, but shall not be subjected to loads or other stresses such as bending or twisting that can damage the entry seals. For level C preservation-packaging, semi-conductors or solid-state devices subject to electromagnetic degradation shall be protected with a wrap of aluminum foil or a barrier bag employing aluminum foil as a laminate of the bag material. Leads and terminals shall be protected as specified herein. Wraps and bags shall be closed by heat sealing.

5.5 Data. Drawings, microfilm, miscellaneous reports, and so forth, to be forwarded to contracting agency in compliance with the requirements of this specification, or as specified in the contract data requirements list, shall be packaged and packed level C in accordance with the applicable data specification or, where no preparation for delivery requirements exist, data shall be prepared in accordance with the level C requirements of MIL-E-17555.

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6. NOTES

6.1 Ordering data.

6.1.1 Procurement requirements. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type and class required (see 1.2).
(c) Quantity of items required.
(d) Probe tube, if required (see 3.6.2).
(e) Hose length, when other than as specified (see 3.6.1.1).
(f) Calibration kit, type and quantity of gas required (see 3.3.8.1 and 3.6.7).
(g) Repair parts required (see 3.9).
(h) Tests conditions other than as specified (see 4.1.2).
(i) Test position of indicator when other than as specified (see 4.5.4).
(j) Preservation, packaging, packing, and marking (see 5.2).

6.1.2 Data requirements. When this specification is used in a procurement which incorporates a DD Form 1423 and invokes the provisions of 7-104.9(n) of the Armed Services Procurement Regulations, the data requirements identified below will be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements List (DD Form 1423) incorporated into the contract. When the provisions of ASPR-7-104.9(n) are not invoked, the data specified below will be delivered by the contractor in accordance with the contract requirements. Deliverable data required by this specification is cited in the following paragraphs:

Paragraph	Data requirement	Applicable DID	Option
3.8.1	Drawings, engineering and associated lists level 3 (production)	DI-E-7015	Type: Detail Design activity designation: By contractor Drawing number: By contractor Delivery of hard copy: To procuring activity
3.8.2	Technical manuals, commercial	DI-M-2050 DI-M-2044	
4.1.1	Inspection systems report	UDI-T-23741	

(Copies of data item descriptions required by the contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

6.2 Caution. The indicators covered by this specification are not for use in oxygen enriched atmospheres (atmospheres containing more than 25 percent oxygen by volume). The oxygen indicators (class 2 units) are for indicating oxygen sufficiency in the range of 0 to 25 percent.

6.3 With respect to products requiring qualification, awards will be made only for products which are at the time set for opening of bids, qualified for inclusion in applicable Qualified Products List QPL 002703 whether or not such products have actually been so listed by that date. The attention of the contractors is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the Qualified Products List is the Naval Ship Engineering Center, Department of the Navy, SEC 6124, Washington, D.C. 20362 and information pertaining to qualification of products may be obtained from that activity. Application for Qualification tests shall be made in accordance with "Provisions Governing Qualification SD-6" (see 6.3.1).

6.3.1 Copies of "Provisions Governing Qualification SD-6" may be obtained upon application to Commanding Officer, Naval Publications and Forms Center, 5801 Tabor Avenue, Philadelphia, Pennsylvania 19120.

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6.4 Sub-contracted material and parts. The preparation for delivery requirements of referenced documents listed in section 2 do not apply when material and parts are procured by the contractor for incorporation into the equipment and lose their separate identity when the equipment is shipped.

6.5 Revision letters are not used to denote changes due to the extensiveness of the changes.

Preparing activity:
Navy - SH

(Project 6665-N383)

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INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

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